Assignment4UseCase.pdf

1. Use Case:

Solve the Sudoku Puzzle

2. Brief Description:

The program is run with one or more puzzle file. The program prints out the name of each puzzle file, the puzzle contained within, how many solutions there are, and the solutions to the puzzle. If the puzzle has no solutions, prints “No solution found”.

3. Step-by-Step flow:
    a. User starts program with or without text file names
    b. System selects files to work through (automatically uses puzzle1.txt - puzzle5.txt if no file inputs)
    c. For each puzzle, reads 81 cells and prints the puzzle and the file name
    d. Searches for solutions and prints each one
    e. Prints how many solutions were found. Prints “No solutions found” if none exist.
    f. Does the next file and ends after the final one.

4. Detailed description:
    - Actor: User
    - Preconditions: There must be an executable, and the puzzle file must either be typed after ./Assignment4 in the command line, or there must be files with the naming convention of puzzle1.txt – puzzle5.txt in the same file.
    - Postconditions: A report has been printed for every file.
    - Main flow: listed in Step-by-Step flow
    - Alternate flows:
        - Multiple solutions: all are printed and numbered
        - Custom file names: the names given on the command line replace the default puzzle1.txt – puzzle5.txt
    - Exceptions:
        - File cannot be opened: prints “Could not open the file” and continues
        - File is not 81 cells: prints error and continues
        - Givens contradict each other, or the search finds no valid fill: prints “No solution found” and continues